task1

September 11, 2026

[2]:
```
import pandas as pd
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

train_data = pd.read_csv("ds1_train.csv")
test_data = pd.read_csv("ds1_test.csv")
valid_data = pd.read_csv("ds1_valid.csv")

X = ["x_1", "x_2"]
X_train = train_data[X]
y_train = train_data["y"]

X_test = test_data[X]
y_test = test_data["y"]

X_valid = valid_data[X]
y_valid = valid_data["y"]
```

[3]:
```
def standard_scaler(X):
    for col in X.columns:
        mean = X[col].mean()
        std = X[col].std()
        X[col] = (X[col] - mean) / std

    return X

X_train_scaled = standard_scaler(X_train.copy())
X_test_scaled = standard_scaler(X_test.copy())
X_valid_scaled = standard_scaler(X_valid.copy())
```

[4]:
```
best_k = 1
best_accuracy = 0
accuracy_scores = []

for k in range(1, 100):
    knn_temp = KNeighborsClassifier(n_neighbors=k)
    knn_temp.fit(X_train_scaled, y_train)
```

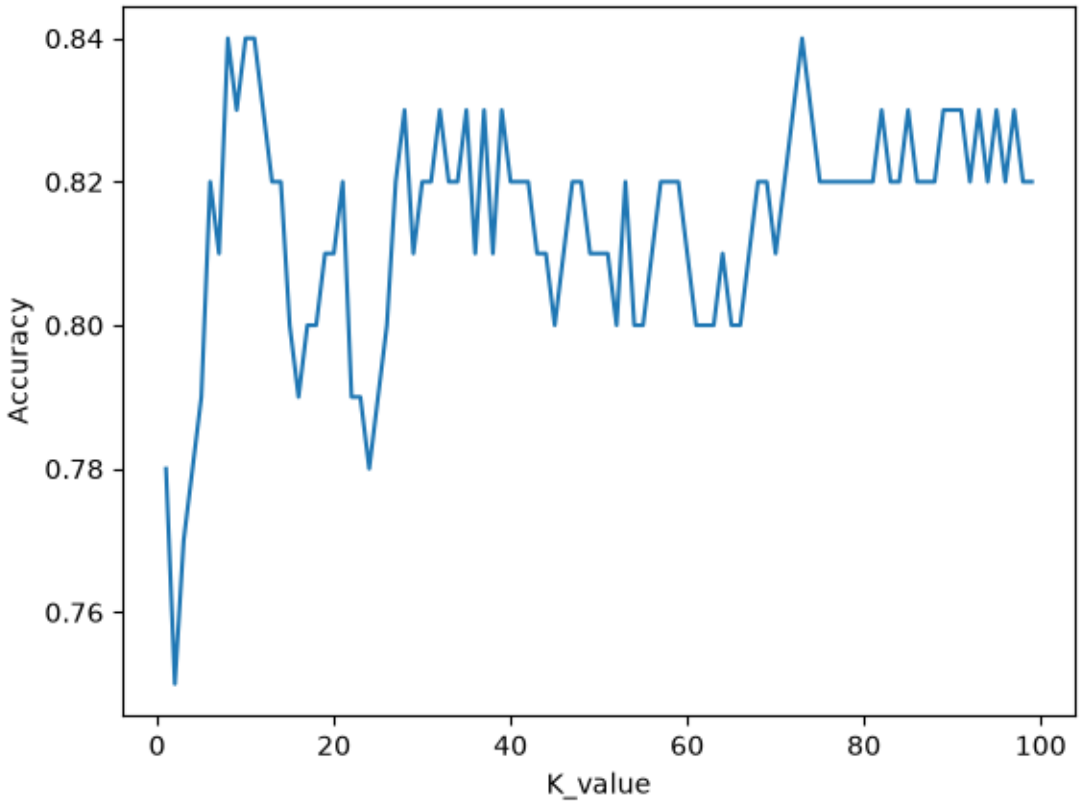
```
    predictions = knn_temp.predict(X_valid_scaled)
    accuracy = accuracy_score(y_true=y_valid, y_pred=predictions)
    accuracy_scores.append(accuracy)
    if accuracy > best_accuracy:
        best_accuracy = accuracy
        best_k = k

print(f"Best k = {best_k} with accuracy = {best_accuracy}")

import matplotlib.pyplot as plt

plt.plot(range(1, 100), accuracy_scores)
plt.xlabel("K_value")
plt.ylabel("Accuracy")
plt.show()
```

```
Best k = 8 with accuracy = 0.84
```



[5]:
```
knn = KNeighborsClassifier(n_neighbors=best_k)
knn.fit(X_train_scaled, y_train)
test_predictions = knn.predict(X_test_scaled)
```

```
accuracy = accuracy_score(y_true=y_test, y_pred=test_predictions)

print(f"accuracy = {accuracy}")
```

```
accuracy = 0.77
```

[7]:
```
results = pd.DataFrame({
    "x_1": X_test["x_1"],
    "x_2": X_test["x_2"],
    "Original": y_test,
    "Predicted": test_predictions
})

results
```

[7]:

	x_1	x_2	Original	Predicted
0	1.637463	4.152545	0.0	0.0
1	1.726565	3.437593	0.0	1.0
2	1.358686	0.757834	0.0	0.0
3	1.620747	4.862745	0.0	1.0
4	0.963280	0.568776	0.0	0.0
..	…	…	…	…
95	2.299773	21.585631	1.0	1.0
96	1.641268	3.810659	1.0	0.0
97	0.574610	3.545178	1.0	0.0
98	1.835419	3.322410	1.0	1.0
99	4.067959	19.918896	1.0	1.0

[100 rows x 4 columns]